Simran Bilkhu

+1-289-400-6136 | bilkhs2@mcmaster.ca

my-profile | simb2 | my-site

Oakville, Ontario, Canada

Education

- **McMaster University** — *September 2024 - April 2026*
  Master's in Statistics — Hamilton, Canada
  - Grade: 11.86/12
  - Thesis [link]
    A Comparative Study of Bayesian Factor Models under PLT and UGLT Identification Constraints
- **University of Toronto** — *April 2024*
  Specialist in Applied Statistics with a minor in Math (Honours Bachelor of Science) — Toronto, Canada
  - Cumulative GPA (final 2 yrs - 10.5 out of 20.5 total credits): 3.76/4.00

Publications

- **Counter-Example to Continuity of Measure in Uncountable Unions** — *Link to Preprint*
  Submitted for review

Projects & Presentations

- **Statistical Society of Canada Annual Meeting** — *June 2, 2026*
  Student Research Poster Presentation Competition (ongoing) — *Link to Project*
  - Scheduled to present thesis research comparing Bayesian factor models under PLT and UGLT identification constraints; submitted as part of the SSC student research competition.
- **Bird Migration Count Forecasting** — *December 2024*
  Assignment — *Link to Project*
  - Forecasted population trends of *Peregrine Falcons* and *Greater Roadrunners* using satellite climate data: modelled counts (13,642 obs, 2001–2005) from time, location, and weather features.
  - A spatio-temporal mixed model was used to reduce test error from 1,377 to 31 vs. ridge-penalised Poisson regression, with warmer temperatures positively associated with counts.
- **Probability and Stochastics** — *July 2025*
  Presentation — *Link to Project*
  - Prepared lectures on martingales, and their relationships to Markov processes, Wiener processes, and Poisson processes.
  - Provided proofs of several theorems and properties.
  - Included an introduction to stochastic kernels, and their connection to stochastic matrices for discrete state spaces.
- **Bayesian Nonparametrics** — *January 2025 - April 2025*
  Assignment — *Link to Project*
  - Provided a thorough proof showing that the proportions from the Chinese Restaurant Process converge in distribution to the GEM distribution.
- **Bayesian Nonparametrics** — *January 2025 - April 2025*
  Assignment — *Link to Project*
  - Showed that the size-biased permutations of the Poisson-Dirichlet distribution is given by the 2-parameter GEM distribution.
- **Statistical Learning** — *January 2025 - April 2025*
  Impacts of Diet Intervention on the Gut Microbiome — *Link to Project*
  - Applied dimension reduction via Principal Component Analysis (PCA) on microbiome count data with over 1400 features, and identified relevant taxa.
  - Used model-based clustering techniques to explore group differences in microbial composition between plant and animal based diets to identify differing microbiome profiles.

◦ Identified important microbial taxa for dietary classification by analyzing loadings from principal components.

- **Research Group Speaker** *April 2025*
  A Review of Sparse Bayesian Factor Analysis when the Number of Factors is Unknown Link to Project
  ◦ Reviewed and presented work by Frühwirth-Schnatter et al., highlighting a Bayesian approach to factor analysis with an unknown number of latent factors using spike-and-slab priors.
  ◦ Provided a background on shrinkage priors, UGLT structures, and *reversible jump MCMC*.

- **Research Group Speaker** *January 2025*
  A Review of Sparse Principal Component Analysis Link to Project
  ◦ Presented key theoretical links between factor analysis and PCA, highlighting the benefits of sparse representations for interpretability in high-dimensional settings.
  ◦ Compared sparse PCA via penalized optimization (see Zou et al.) to varimax rotation, using simulation studies to assess interpretability and mean squared error across varying sparsity levels.

- **Data Science** *November 2024*
  Establishing Baseline Climate Partitions via Clustering Link to Project
  ◦ Analyzed time-series raster climate data (temperature, precipitation, elevation) across Canada using python, focusing on spatial patterns and long-term climatic trends.
  ◦ Performed K-means and Agglomerative clustering methods to group geographic regions using silhouette analysis to choose the number of clusters.
  ◦ Visualized and interpreted clustering results through geographic heatmaps, and cluster-wise summaries.

- **Data Science** *September 2024*
  Expansion of Canadian Agriculture and the Relationship with Wetlands in Boreal Shield Provinces Link to Project
  ◦ Conducted geospatial analysis of 66 GeoTIFF satellite land cover maps (2013–2023) using R's terra package to quantify agricultural land use and wetland area.
  ◦ Preprocessed raster data into a tidy format and scaled estimates to square kilometers for year-by-year provincial comparison.
  ◦ Designed and deployed a Shiny web app for interactive visualization, allowing users to explore province-specific trends and relationships across time (*link to site*).
  ◦ Addressed limitations in remote sensing data such as non-random missingness due to satellite imaging challenges, and discussed implications for ecological monitoring and policy.

- **Research Group Speaker** *September 2024*
  Hamiltonian Monte Carlo Link to Project
  ◦ Presented a conceptual and mathematical overview of Hamiltonian Monte Carlo (HMC), explaining the role of Hamiltonian dynamics.
  ◦ Co-developed a Shiny web app allowing users to simulate from normal and beta distributions using HMC

Work Experience

- **McMaster University**, Research Assistant II *May 2026 - Present*
- **McMaster University**, Intermediate Probability Theory (TA) *September 2025 - December 2025*
- **McMaster University**, Statistical Methods and Applications (TA) *January 2025 - April 2025*
- **McMaster University**, Linear Regression Models Using SAS (TA) *September 2024 - December 2024*
- **Sodexo Canada**, IT Intern *Mar 2020 - Aug 2022*

Works Cited

Frühwirth-Schnatter, Sylvia, et al. "Sparse Bayesian Factor Analysis when the number of factors is unknown (with Discussion)". *Bayesian Analysis*, vol. 20, no. 1, 2025, pp. 213–344.

Zou, Hui, et al. "Sparse principal component analysis". *Journal of computational and graphical statistics*, vol. 15, no. 2, 2006, pp. 265–86.